

NTRK3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK3 gene fusions from the msk_impact_50k_2026 study, covering 58 fusion events. 55/58 fusions (94.8%) were in-frame. 55/58 fusions (94.8%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0695489$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK3.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 58 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ETV6, observed in 55 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions retained the domain, $p=0.0695489$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: NTRK3 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 57 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (permutation-corrected $p=0.036963$, statistically significant at $\alpha=0.05$). This is -5 aa from the nearest configured domain boundary at 160 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: NTRK3 has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: NTRK3 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
ETV6	55	0.9482758620689655	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.47798684709322253	1
DOT1L	1	0.017241379310344827	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2932577830538136	2
ZNF710	1	0.017241379310344827	0.20043213737826426		0.11159235668789813		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10288471662979178	3
VPS39	1	0.017241379310344827	0.20043213737826426				['domain_retention', 'recurrence']	0.10050990570485364	4

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
155	0	55	1	1	0.0	0.03508771929824562	1.4548448600085102
466	14	41	2	0	0.0	0.07518796992481205	1.1238516409670856
472	15	40	2	0	0.0	0.08521303258145364	1.069493978644493

2.3 domain_retention frame_domain_contingency_table

54	1
1	1

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein tyrosine and serine/threonine kinase	57	55	0	2	0.0

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
DOT1L	1
ETV6	55
VPS39	1
ZNF710	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	retained	score	is_fusion	description	disrupted_domain	disrupted_domain	description_annotation	effect_on_fusion
EVT-P-0013664-T02-IM5-1	P-0013664-T02-IM5		ETV6-NTRK3	ETV6	in-frame	three_prime	88558191	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q26.1)(chr12:g.12032077: chr15:g.12033175) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0061265-T01-IM7-2	P-0061265-T01-IM7		ETV6-NTRK3	ETV6	in-frame	three_prime	88554122	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion: t(12;15)(p13.2;q25.3)(chr12:g.12033175: chr15:g.12033175) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0058408-T01-IM6-3	P-0058408-T01-IM6		ETV6-NTRK3	ETV6	in-frame	three_prime	88492156	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion: t(12;15)(p13.2;q25.3)(chr12:g.12035760: chr15:g.12033175) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0056156-T02-IM7-4	P-0056156-T02-IM7		ETV6-NTRK3	ETV6	in-frame	three_prime	88624274	12	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion: t(12;15)(p13.2;q25.3)(chr12:g.12011957: chr15:g.12033175) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0055703-T01-IM6-5	P-0055703-T01-IM6		ETV6-NTRK3	ETV6	in-frame	three_prime	88505980	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion: t(12;15)(p13.2;q25.3)(chr12:g.12024785: chr15:g.12033175) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	chr1	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0010668-T01-IM5-22	P-0010668-T01-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88513282	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109:	NA	
EVT-P-0014872-T02-IM6-23	P-0014872-T02-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88548029	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946:	NA	
EVT-P-0014872-T03-IM6-24	P-0014872-T03-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88548029	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946:	NA	
EVT-P-0013196-T04-IM6-25	P-0013196-T04-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072:	NA	
EVT-P-0016507-T01-IM6-26	P-0016507-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709:	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_stat	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein	breakpoint_4	breakpoint_5	main_ref	chr	start	end	domain	disrupted_domain	description	annotation	effect_on_fra	
EVT-P-0026390-T01-IM6-37	P-0026390-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88651131	12	466	True	retained	1.0	False						Protein tyrosine and serine/threonine kinase	ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20):t(12;15)(p13.2;q25.3)(chr12:g.120073037: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3}	NA	
EVT-P-0026696-T01-IM6-38	P-0026696-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88657925	12	466	True	retained	1.0	False						Protein tyrosine and serine/threonine kinase	ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20):t(12;15)(p13.2;q25.3)(chr12:g.120073037: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3}	NA	
EVT-P-0027006-T01-IM6-39	P-0027006-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	retained	1.0	False						Protein tyrosine and serine/threonine kinase	ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3}	NA	
EVT-P-0027006-T03-IM6-40	P-0027006-T03-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	retained	1.0	False						Protein tyrosine and serine/threonine kinase	ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3}	NA	
EVT-P-0027006-T04-IM6-41	P-0027006-T04-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	retained	1.0	False						Protein tyrosine and serine/threonine kinase	ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352: Protein Fusion: in frame {ETV6:NTRK3}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	chr1	start	end	domain	disrupted	description	annotation	effect_on_fra
EVT-P-0007911-T02-IM5-47	P-0007911-T02-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88592430	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12007142: Protein fusion: in frame (ETV6-NTRK3)	NA (2338)	
EVT-P-0033921-T03-IM6-48	P-0033921-T03-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88516534	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161: Protein Fusion: in frame {ETV6:NTRK3}	NA (2338)	
EVT-P-0007772-T03-IM5-49	P-0007772-T03-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88600844	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257:) Protein fusion: in frame (ETV6-NTRK3)	NA (2338)	
EVT-P-0007772-T02-IM5-50	P-0007772-T02-IM5		NTRK3-ETV6	ETV6	in-frame	five_prime	88600844	13	466	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257:) Protein fusion: in frame (NTRK3-ETV6)	NA (2338)	
EVT-P-0035552-T01-IM6-51	P-0035552-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88651251	12	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12020963: Protein Fusion: in frame {ETV6:NTRK3}	NA (2338)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	domain_ref	domain_1	domain_2	domain_3	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0035767-T01-IM6-52	P-0035767-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88505787	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143: Protein Fusion: in frame {ETV6:NTRK3} ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3)	NA	
EVT-P-0007772-T01-IM5-53	P-0007772-T01-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88600844	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3)	NA	
EVT-P-0006480-T01-IM5-54	P-0006480-T01-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88498145	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3)	NA	
EVT-P-0002978-T04-IM5-55	P-0002978-T04-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3) ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3)	NA	
EVT-P-0002978-T05-IM6-56	P-0002978-T05-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576: Protein fusion: in frame (ETV6-NTRK3)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	chr1	start	end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0004737-T01-IM5-57	P-0004737-T01-IM5		NTRK3-ETV6	ETV6	in-frame	three_prime	88632786	12	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12006809:	NA	
EVT-P-0023257-T01-IM6-58	P-0023257-T01-IM6		NTRK3-ETV6	ETV6	in-frame	three_prime	88662190	12	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998:	NA	
EVT-P-0010556-T01-IM5-70	P-0010556-T01-IM5		NTRK3-ZNF710	ZNF710	unknown	three_prime	88576258	13	472	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			{ETV6:NTRK3} NTRK3 (NM_001012338) - ZNF710 (NM_198526) rearrangement: c.-28-10227:ZNF710_c.1415:NTRK3inv	NA	NTRK3inv

3 Figures

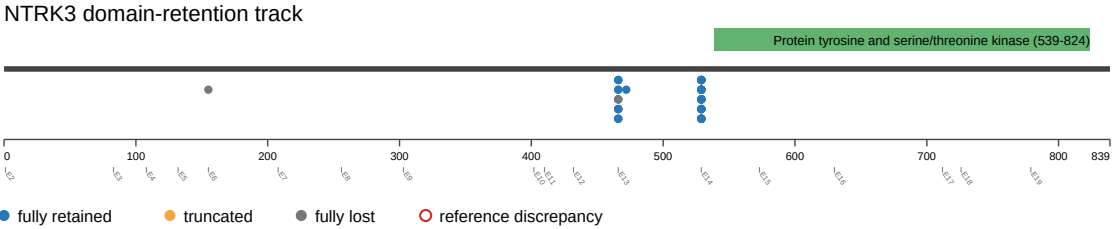


Figure 1: domain retention outliers

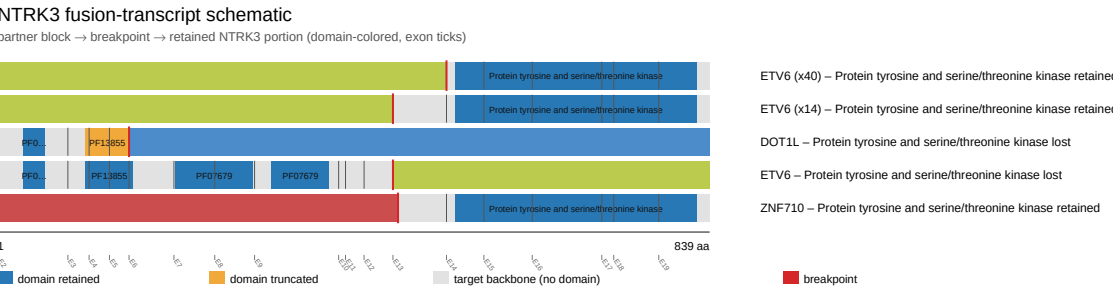


Figure 2: fusion schematic

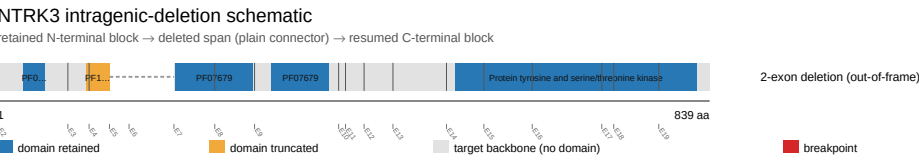


Figure 3: intragenic deletion schematic

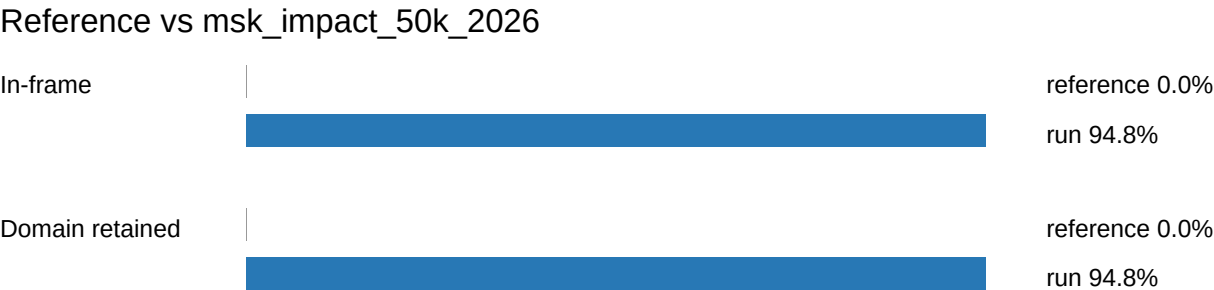


Figure 4: reference comparison